

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

Correct Answer: 14

Rationale

The correct answer is **14**. Let x represent the first integer and y represent the second integer. If the first integer is **5** greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is **462**; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting **462** from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or **-924**, and whose sum is **5**. The two values whose product is **-924** and whose sum is **5** are **33** and **-28**. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting **33** from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by **2** yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding **14** to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting **14** for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by **14** yields $x = 33$. Therefore, the two integers are **14** and **33**, so the smaller of the two integers is **14**.